Week 2 – STP & EtherChannel Lab Report

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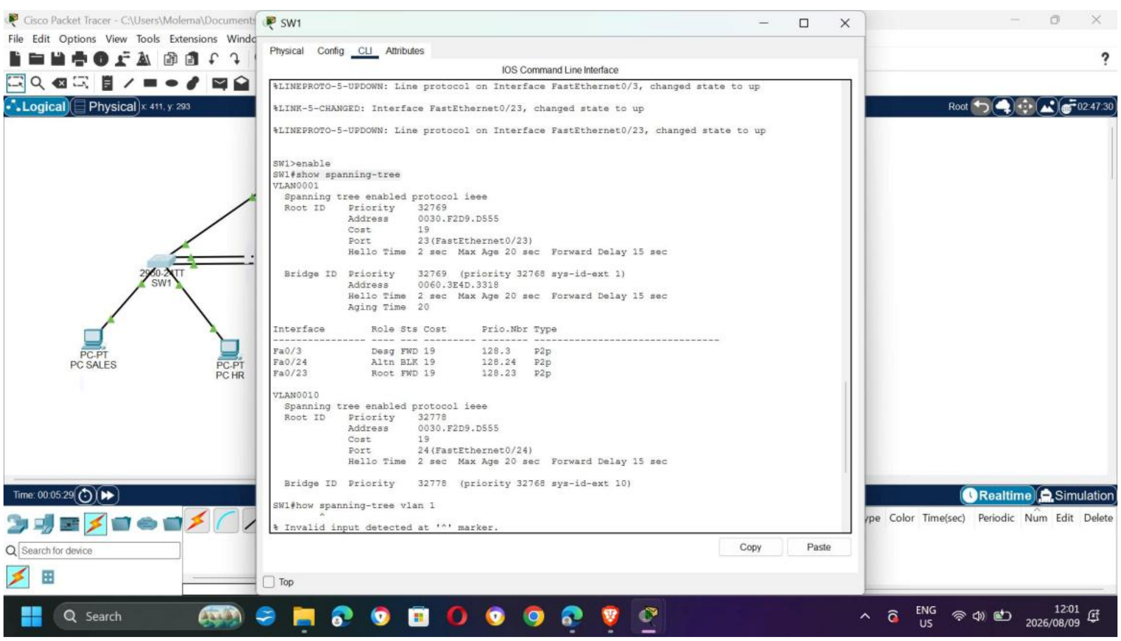
Date: 2026/08/09
Topology: R1 - SW1 - SW2 with 3 PCs (VLAN 10,20,30) and redundant links Fa0/23-24 bundled as LACP

Step 1: Observe Default STP Behaviour

14. Command: show spanning-tree
15. Identify root bridge and blocking port

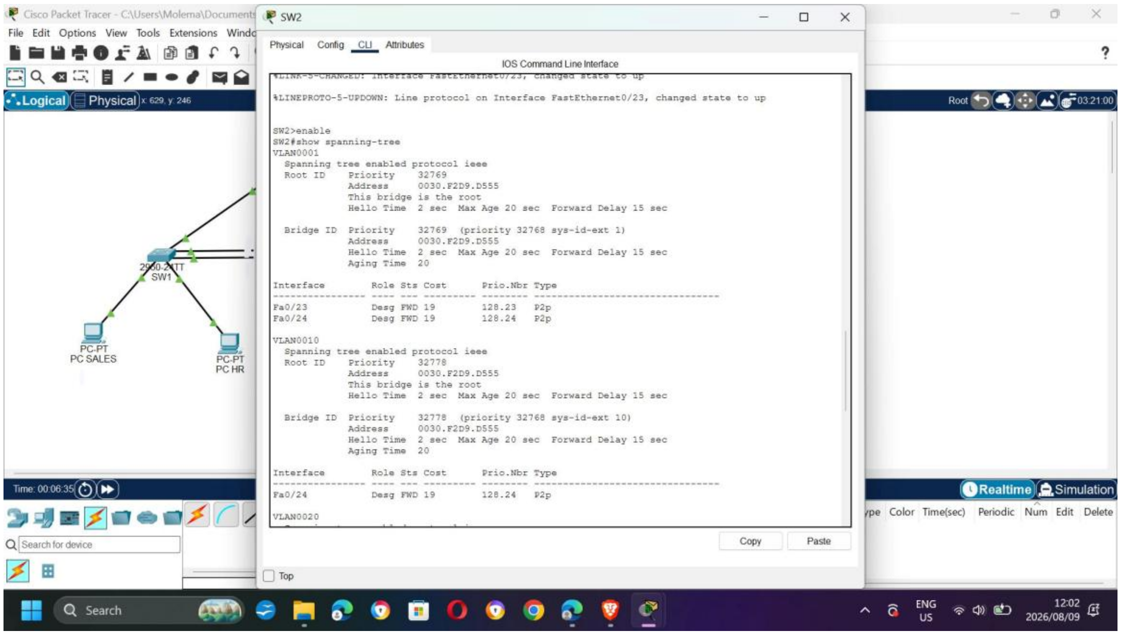
BEFORE changing priority – SW1

Default priority is 32769 (32768 + sys-id-ext 1). The switch with the lowest MAC becomes root. In this lab SW2 was root by default.
From screenshot: Root ID 0030.F2D9.D555 (SW2), Fa0/24 Altn BLK, Fa0/23 Root FWD. Blocking port is Fa0/24 on SW1.



BEFORE changing priority – SW2

SW2 shows "This bridge is the root" – confirms SW2 was default root before tuning.

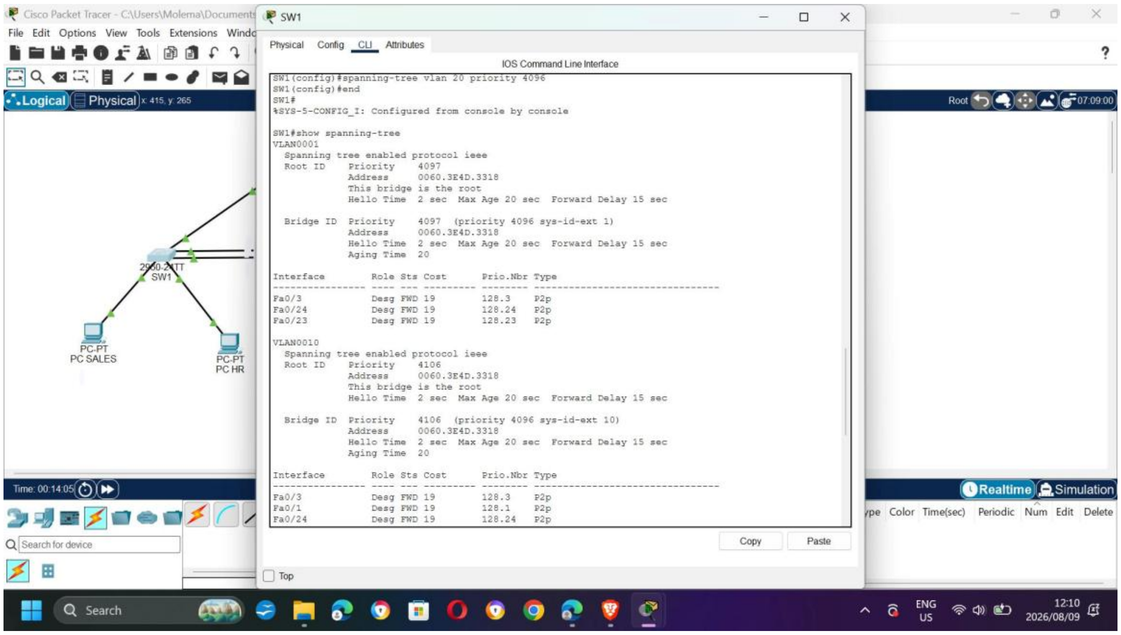


Step 2: Tune STP – Force SW1 as Root Bridge

16. Commands used on SW1:
configure terminal
spanning-tree vlan 1 priority 4096
spanning-tree vlan 10 priority 4096
spanning-tree vlan 20 priority 4096
spanning-tree vlan 30 priority 4096
end
show spanning-tree

AFTER changing priority – SW1 is now Root

After change: Priority 4097 (4096 + 1), Address 0060.3E4D.3318, "This bridge is the root". All interfaces Desg FWD. No BLK on root bridge.



Step 2 – Continued: Convergence Test

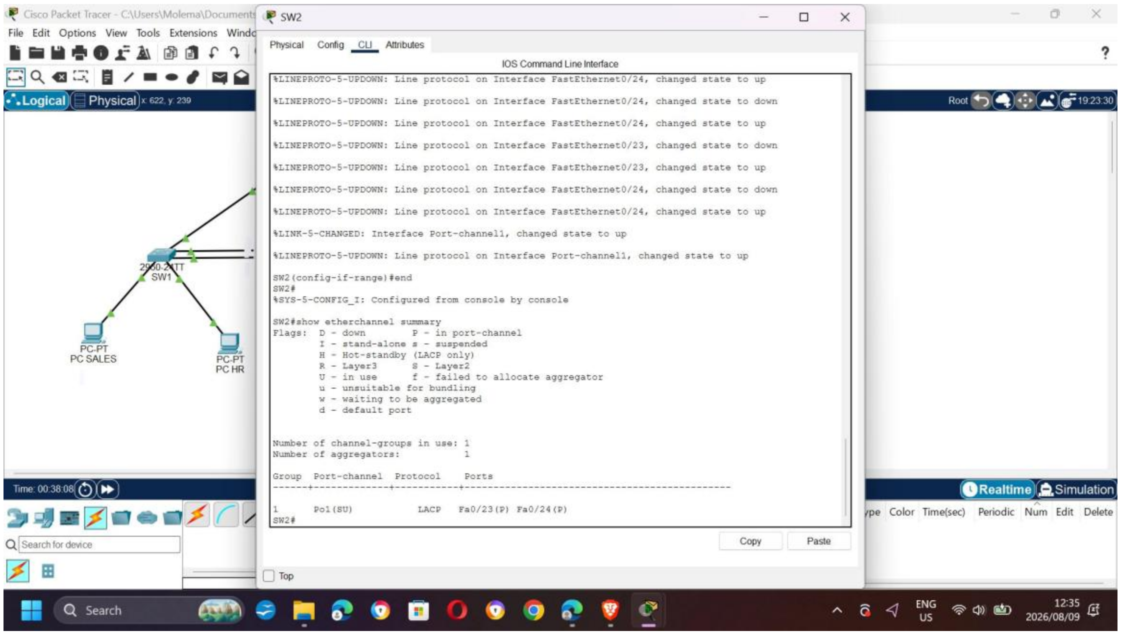
17. Shut down the active forwarding link between SW1-SW2:
interface fa0/23
 shutdown
Result: The previously blocked port (Fa0/24) transitioned BLK -> LIS -> LRN -> FWD.
Measured time: ~30-50 seconds with classic PVST (15 sec Listening + 15 sec Learning).
With RSTP/rapid-pvst it would be ~2-6 seconds.
18. Re-enable: no shutdown on Fa0/23

Step 3: Replace Redundant Links with EtherChannel (LACP)

19. Commands on BOTH SW1 and SW2:
interface range fa0/23-24
switchport mode trunk
channel-group 1 mode active
no shutdown
Note: switchport mode trunk was added to fix DTP mismatch error %EC-5-CANNOT_BUNDLE2

Verification – show etherchannel summary

20. Expected output: Po1(SU) LACP with both ports P (bundled).
This replaces two separate STP links with one logical Port-channel. STP now sees Po1 as a single link, so no port is blocked and bandwidth is 2 Gbps.



SW2
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
SW2(config-if-range)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
SW2#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 1
Number of aggregators: 1
Group Port-channel Protocol Ports
1 Po1(SU) LACP Fa0/23(P) Fa0/24(P)
SW2#
Copy
Paste
Top
PC-PT
PC SALES
PC-PT
PC HR
SW1
Logical
Physical
Realtime
Simulation